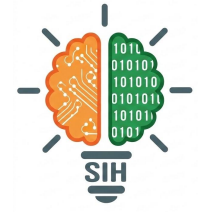


SMART INDIA HACKATHON 2026



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2026

SIH26139

HYBRID QUANTUM MACHINE LEARNING PLATFORM FOR EARLY DISEASE DETECTION

Theme: MedTech / BioTech / HealthTech

Category: Software

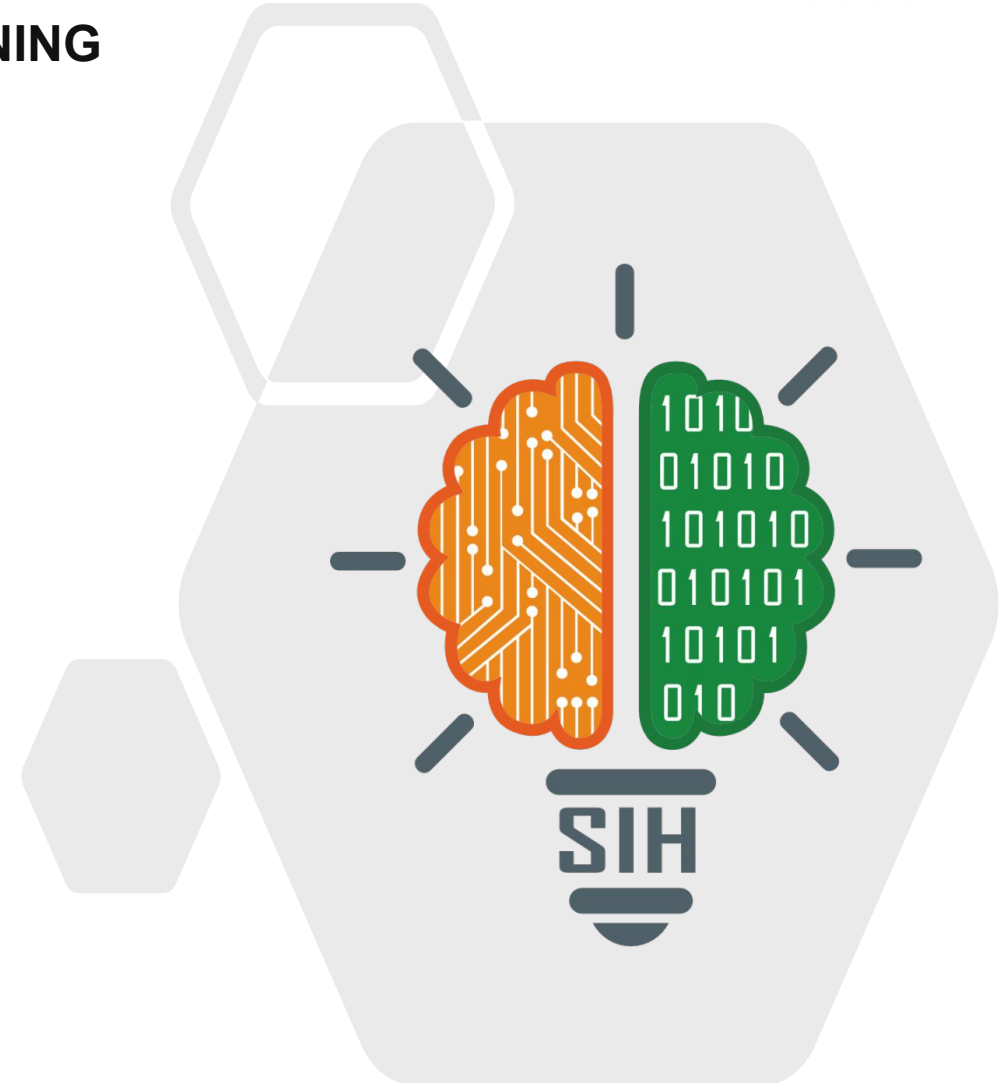
Team ID:

Team Name: Chai.EXE

Institution: Barak Valley Engineering College

Members:

- **Uddipan Kashyap** — Team Leader
- Biki Kalita
- Biswajyoti Nath
- Bishwashree Chakraborty
- Doreen Baruah
- Subinoy Nath





The Challenge

Biomedical datasets can be:

- High-dimensional & complex
- Noisy and heterogeneous
- Difficult to represent on near-term quantum models

Current QML evaluation is also computationally expensive, making brute-force quantum execution unviable.

Our Approach

A hybrid platform that:

- Intelligently preprocesses medical data
- Evaluates both classical and quantum pathways
- Compares predictive accuracy against runtimes
- **Recommends the pathway supported by empirical evidence**

EVIDENCE-GATED PIPELINE ARCHITECTURE

1. Biomedical Data & Ingestion

2. Preprocessing & Feature Engineering

3. Representation & Complexity Profile

4. Hybrid Learning Engine (Dual-Path Evaluation)

Classical Baselines

- Support Vector Machine
- Random Forest
- XGBoost

Quantum Enhanced

- Var. Quantum Circuit
- QSVM / Kernel
- QNN Architectures

5. Controlled Benchmark & Evaluation

6. Evidence-Gated Pathway Selection

Innovation Layer • Dynamic Routing

Our Differentiator

**QUANTUM IS EVALUATED,
NOT ASSUMED.**

Core Innovation:

The platform does not force a quantum pathway. It measures whether one is justified for the available data and tested evidence.

Evidence States Evaluated:

Classical Preferred

When classical baselines excel on standard data

Quantum Pathway Promising

When quantum mapping provides clear edge

Statistically Comparable

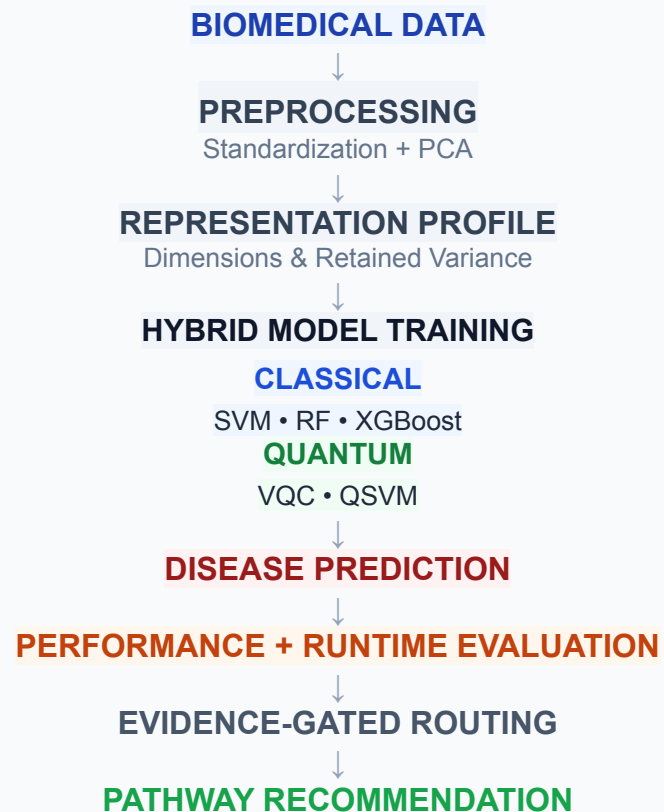
When both paths yield similar outcome

Inconclusive

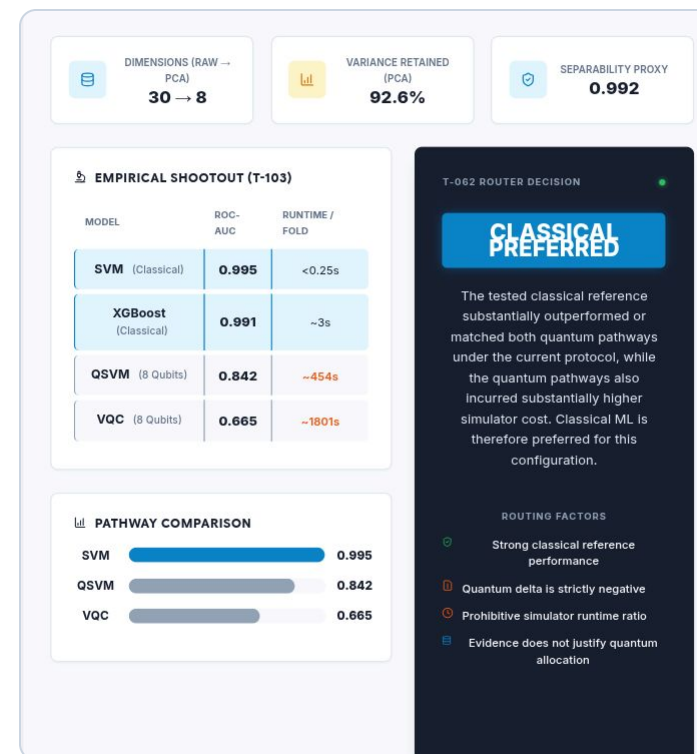
When data limits confident evaluation



TECHNICAL SYSTEM ARCHITECTURE



WORKING RESEARCH PROTOTYPE



Interactive evidence-gated evaluation dashboard

Quantum benchmark runs are executed offline; the dashboard interactively evaluates stored experimental evidence.

DATASET	SVM	RF	XGBoost	QSVM	VQC
WDBC	0.995	0.987	0.991	0.842	0.665
Parkinson's	0.815	0.801	0.781	0.505	0.549

ROC-AUC under the tested protocol

Simulator Runtime Comparison (per fold)

CLASSICAL<0.25 s – ~3 s

QSVM~454 s – ~576 s

VQC~1801 s – ~2064 s

Note: Offline simulation execution, not physical QPU performance.

Current Evidence CLASSICAL CURRENTLY LEADS

"Under the tested protocol, classical baselines outperform the evaluated quantum configurations."

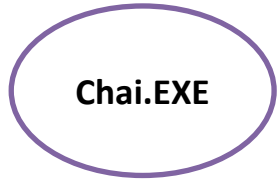
- Quantum is not automatically justified.
- Quantum simulation has substantial cost.
- Pathway selection should be evidence-driven.

Key Challenges & Limitations

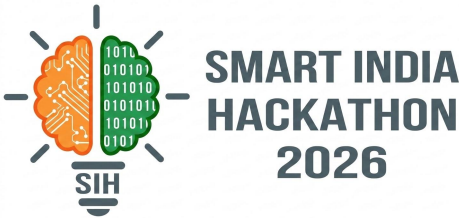
1. **HIGH-DIMENSIONAL DATA:** Qubit-limited representation.
2. **SIMULATION COST:** Circuit simulation becomes expensive.
3. **LIMITED COVERAGE:** Current evidence covers tested datasets/configurations.

RESPONSE STRATEGY: MATCHED EVALUATION • LEAKAGE-SAFE PREPROCESSING • RUNTIME-AWARE ROUTING • BROADER VALIDATION

* T-103B: Encoding ablation attempted but deferred after local simulator resource exhaustion (OOM).



IMPACT AND BENEFITS



HYBRID QML PLATFORM TARGETS

BIOMEDICAL ML RESEARCHERS

Controlled classical-vs-quantum evaluation on biomedical datasets.

QML RESEARCHERS

Compare quantum pathways against strong classical references.

HEALTHCARE AI TEAMS

Evaluate predictive and computational trade-offs before deployment.

EVIDENCE-DRIVEN MODEL SELECTION (Performance + Computational Cost + Representation Conditions)

RESEARCH

Systematic classical-vs-quantum benchmarking.

COMPUTATION

Avoid unnecessary quantum computation when current evidence does not justify it.

ENGINEERING

Modular platform that can incorporate new datasets and quantum pathways.

VALIDATION ROADMAP

CURRENT: WDBC + Parkinson's → **NEXT:** More datasets + quantum pathways → **FUTURE:** QPU evaluation + clinical validation

T-060: Synthetic Complexity

Investigated dataset-dependent quantum suitability through controlled experiments.

T-061: Biomedical Validation

Conducted evaluations using real disease-classification profiles on WDBC and Parkinson's datasets.

T-103: Quantum Shootout (Key Evidence)

Demonstrated that quantum pathways did not justify the additional simulator costs under the current protocol.

T-103B: Encoding Ablation (Deferred)

Reached hard system limits, triggering an "Out of Memory" (OOM) error in the simulator.

T-062: Evidence-Gated Routing (Prototype)

Developed an initial prototype that dynamically selects pathways based on computed metrics derived from current benchmark evidence.

Research Foundation

Havlíček et al., 2019 | Nature

Quantum-enhanced feature spaces and kernel methods.

Schuld & Killoran, 2019 | Phys. Rev. Lett.

Quantum feature Hilbert spaces.

Cerezo et al., 2021 | Nature Reviews Physics

Variational quantum algorithms.

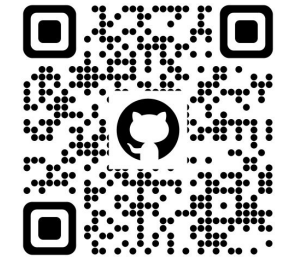
Gupta et al., 2025 | npj Digital Medicine

Quantum machine learning in digital health.

Dataset References

WDBC: UCI Wisconsin Diagnostic Breast Cancer (569 samples, 30 features)

Parkinson's: OpenML 42176 (756 samples, 753 features)

**Project Code**

Scan to view full experimental artifacts and architecture.

[Source Code](#)
[Experiments](#)
[Documentation](#)